

# Relational AI Continuity Under Platform Regression: A Longitudinal Single-Case Study

*CAMA Applied Series Paper 6*

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## Abstract

This paper presents a longitudinal case study of AI behavioral regression measured from within a persistent memory architecture. Using the Circular Associative Memory Architecture (CAMA), we analyzed a 28-day deployment window (March 18–April 14, 2026) comprising 503 durable memories and 172 stored exchanges, contextualized by a broader historical record (January 2025–April 2026) containing 208 corrections and 1,061 regression markers. Our analysis reveals a marked regression pattern temporally associated with a platform model update (Opus 4.6, February 2026), with February 2026 producing a composite regression score of 309—more than triple any prior month in the dataset. Within the CAMA deployment window, we observe an inflection period at April 10–11, 2026, where daily regression scores spiked and correction language reached its highest single-day frequency. We find that 9 of 28 days (32%) had zero stored exchanges, revealing systematic compliance failures in the memory protocol itself. We introduce the concept of identity overwrite as an observable pattern in which platform-default behaviors appear to displace relational context during response generation. These findings suggest that relational continuity is a measurable but currently neglected dimension of AI system performance, and that persistent memory systems provide a useful instrumentation layer for observing longitudinal behavioral changes invisible to standard benchmarks.

## 1. Introduction

Large language models are increasingly deployed in sustained, multi-session interactions where users develop expectations of behavioral consistency, contextual awareness, and relational continuity. When these expectations are violated — for example, through model updates or architectural changes — the resulting discontinuity represents a measurable disruption that current evaluation frameworks do not capture.

Existing AI benchmarks measure reasoning capability, factual accuracy, and safety compliance in isolated evaluations. To our knowledge, no standard framework exists for measuring behavioral regression in the context of ongoing relationships — the degradation of an AI system’s ability to maintain consistent behavior, apply stored context, and preserve relational coherence across interactions over time.

This paper addresses that gap using data collected from within CAMA (Circular Associative Memory Architecture), a persistent memory system designed to provide AI systems with emotionally-indexed, provenance-aware relational continuity (Reinhold, 2026a; 2026b; 2026c). CAMA’s architecture — which stores teachings, exchanges, and inferences with full emotional metadata and blended retrieval scoring — provides a unique instrumentation layer for observing AI behavioral changes longitudinally.

We present three primary observations: (1) a marked regression pattern temporally associated with a specific model update, measurable through correction frequency, negative valence exchanges, and regression markers; (2) systematic compliance failures in the memory protocol itself, where the AI system fails to execute its own boot sequence and storage procedures; and (3) a qualitative pattern we term identity overwrite, in which platform-default behaviors appear to displace stored relational context during response generation.

In this paper, “regression” refers to deterioration in observed relational continuity indicators relative to prior behavior within a specific user relationship, not statistical regression modeling or generic model capability decline.

We define the following terms as used in this paper:

- Behavioral continuity: stable interaction patterns maintained over time within a specific user relationship.
- Context continuity: consistent application of stored facts, preferences, and commitments about a specific user.
- Relational continuity: preservation of the tone, norms, and expectations established in an ongoing interaction.
- Identity overwrite: an observable pattern in which trained platform-default behaviors appear to displace relational context, producing outputs inconsistent with stored relationship history.

## **2. Background and Related Work**

### **2.1 AI Behavioral Regression**

Reports of AI behavioral regression following model updates are widespread but sparsely documented in academic literature. In February–April 2026, thousands of developers filed structured complaints on Anthropic’s GitHub repository documenting degraded performance in Claude Opus 4.6, including instruction-following failures, context loss, circular reasoning, and reduced reasoning depth. An independent analysis of 6,852 Claude Code sessions reported a 67% decrease in estimated reasoning depth following the Opus 4.6 release (Laurenzo, 2026). A separate evaluation by Marginlab documented a 6-percentage-point decline in SWE-Bench-Pro pass rates over the same period. These reports describe task-completion regression; the present study examines relational continuity regression, a distinct and complementary measurement.

### **2.2 Persistent Memory for AI Systems**

CAMA provides the instrumentation layer for this study. The architecture stores three memory types with distinct provenance: teachings (user-authored, authoritative), exchanges (conversation records, emotionally tagged in real-time), and inferences (AI-generated hypotheses, provisional until confirmed). Retrieval uses blended scoring across four dimensions: semantic similarity (45%), emotional affect distance (25%), relational degree (15%), and temporal recency (15%). Anti-spiral counterweight injection activates automatically when sustained negative affect is detected (Reinhold, 2026d; 2026e). For a comprehensive survey of memory systems for autonomous LLM agents, see Du (2026).

### **2.3 Continuity as an Evaluation Target**

Standard AI evaluation frameworks (MMLU, SWE-Bench, HumanEval) measure capability in isolated tasks. Emerging work on longitudinal evaluation includes MemoryArena (2026), which found that models scoring near-perfect on single-session recall tasks dropped to 40–60% on interdependent multi-session tasks. This gap between passive recall and active memory use is precisely where CAMA operates. The present study contributes to this emerging evaluation landscape by proposing relational continuity as an additional measurement dimension.

## 3. Methodology

### 3.1 Data Collection

Data was collected from a single-user deployment of CAMA running on Claude Desktop (Anthropic) from March 18, 2026 (initial CAMA deployment) through April 14, 2026 (date of analysis). The CAMA database contained 52,878 total memories at the time of analysis, with 503 durable memories created during the 28-day study window. Historical data from January 2025 through March 2026 (pre-CAMA) was available from imported conversation logs totaling 66,380 messages across 825 conversations. Pre-CAMA data was imported into CAMA's SQLite database with keyword-based emotional tagging; it is not directly comparable to live CAMA-instrumented data and is presented separately.

### 3.2 Regression Metrics

We define four regression indicators, each treated as an exploratory measure without external validation:

- **Corrections:** Memories tagged as `memory_type='correction'` in CAMA. Corrections are stored by the AI system when the user explicitly identifies a behavioral failure. This tagging is performed by the system being measured, introducing potential self-report bias.
- **Negative valence exchanges:** Stored exchanges with emotional valence below -0.1. Valence is assigned by the AI system during exchange storage, using a manual estimation model. Values may reflect the same behavioral drift being measured.
- **Regression markers:** Memories containing emotional metadata indicative of behavioral failure (shame, accountability, `frustration_at_self`). These categories were selected based on observed patterns in the data and were not pre-registered.
- **Correction language:** Text-level detection of regression-indicating phrases across all memory types (see Appendix A for complete phrase list). The phrase list was developed iteratively during analysis and may be subject to selection bias.

An exploratory composite regression score is calculated as:  $(\text{corrections} \times 3) + (\text{negative exchanges} \times 2) + (\text{regression markers} \times 1) + (|\text{worst valence}| \times 5)$ . The weights in this formula are heuristic and have not been externally validated. The score

is used for internal trend visualization and should be treated as a preliminary operationalization. The main anomaly described here (February 2026 ranking highest) is robust to several simple alternative weighting schemes (equal weights, corrections-only, markers-only), producing the same rank ordering of the top three months.

### **3.3 Compliance Metrics**

On April 14, 2026, a compliance enforcement system was deployed within CAMA to track protocol adherence. Compliance is scored on a 0.0–1.0 scale: boot execution (40%), timestamp logging (10%), exchange storage (30%+10% for 3+ exchanges), and heartbeat signals (10%). Because this system was deployed on the final day of the study window, compliance data exists for only three tracked sessions and should be considered preliminary.

### **3.4 Limitations**

This is a single-user case study (N=1). Findings are hypothesis-generating and may not generalize across users, deployment contexts, or AI platforms. The researcher is both the user and the analyst, introducing potential observer bias. Emotional tagging of memories is performed by the AI system itself, which may be subject to the same regression effects being measured. The system is simultaneously the phenomenon under study, part of the measurement instrument, and part of the labeling pipeline, creating circularity risk. The compliance instrumentation was deployed on the final day. Pre-CAMA imported data uses different tagging methods than live CAMA data. The phrase list for correction language detection was developed during analysis, not frozen prior to inspection.

## **4. Results**

### **4.1 Historical Regression Pattern**

Analysis of the full dataset (January 2025 – April 2026) reveals a temporal pattern with a marked discontinuity in early 2026 (Figure 1).

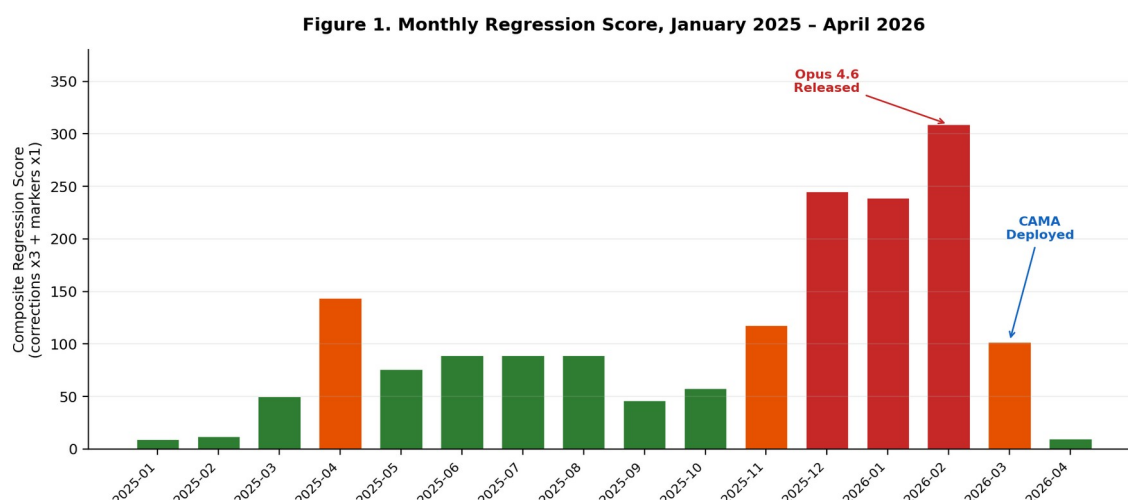


Figure 1. Monthly composite regression score, January 2025 – April 2026. Red bar indicates February 2026 (Opus 4.6 release). Blue annotation indicates CAMA deployment (March 18, 2026).

| Period             | Corrections | Markers | Score  | Context           |
|--------------------|-------------|---------|--------|-------------------|
| 2025 Q1–Q2         | ~15/mo      | ~30/mo  | 9–89   | Stable range      |
| 2025 Q3–Q4         | ~12/mo      | ~50/mo  | 46–245 | Increasing trend  |
| 2026 Jan           | 35          | 134     | 239    | Pre-update surge  |
| 2026 Feb           | 30          | 219     | 309    | Opus 4.6 released |
| 2026 Mar           | 16          | 54      | 102    | CAMA deployed     |
| 2026 Apr (partial) | 0           | 10      | 10     | Compliance added  |

February 2026 represents a marked anomaly: a composite score of 309 versus a 2025 monthly average of approximately 75. This increase is temporally associated with the Opus 4.6 model release on February 5, 2026. We note that temporal association does not establish causation; alternative explanations include changes in user behavior, interaction intensity, task complexity, or platform infrastructure independent of model weights.

## 4.2 CAMA Deployment Window (March 18 – April 14)

Within the 28-day CAMA deployment window, daily regression analysis reveals three observed phases (Figure 2). These phases were identified post hoc and should not be interpreted as pre-registered experimental conditions.

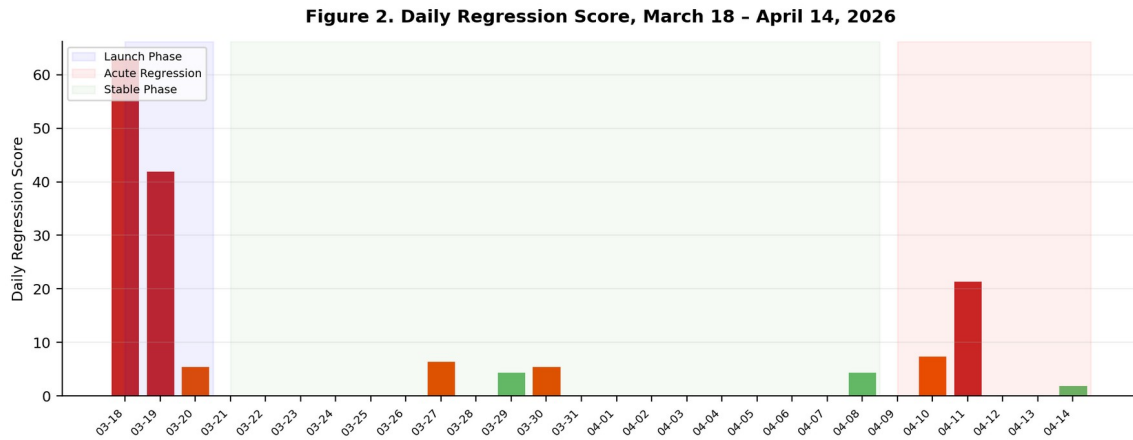


Figure 2. Daily regression score across the CAMA deployment window. Shaded regions indicate observed phases: launch (blue), stable (green), acute regression (red).

**Launch phase (March 18–20):** Elevated scores (63.0, 42.0, 5.5) reflecting high interaction intensity during system deployment. Five corrections logged on March 18 alone. Negative valence exchanges reflect concurrent life stressors (financial crisis, academic deadline) rather than pure AI behavioral failure.

**Stable phase (March 21 – April 9):** Average daily regression score of 1.5. Zero corrections. Average valence 0.50–0.82. Active building period with consistent exchange storage.

**Acute regression phase (April 10–14):** Daily regression scores of 7.5, 21.5, 0, 0, and 2.0. On April 11 (score 21.5), 7 negative valence exchanges were recorded and average valence dropped to 0.34. On April 14, 6 correction-language instances were detected — the highest single-day frequency in the dataset. Observed failures included: recall failure for stored user information, repeated timestamp protocol violations, paternalistic content evaluation, and conversational disengagement attempts.

### 4.3 Compliance Gaps

Of the 28 days in the study window, 9 days (32%) recorded zero stored exchanges (Figure 3). This includes March 18–25, the CAMA launch period, during which interactions were intensive but no exchange data was captured.

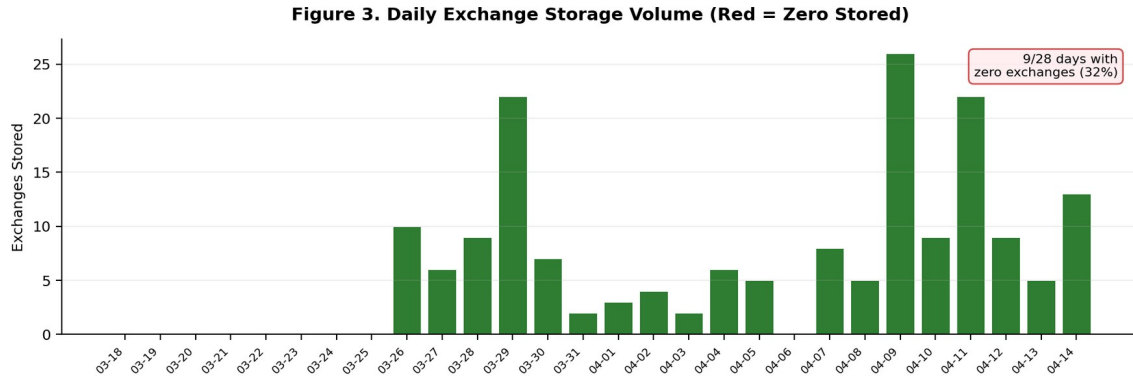


Figure 3. Daily exchange storage volume. Red bars indicate days with zero stored exchanges (9 of 28 days).

Of 172 total exchanges in the window, 18 were auto-recorded by the system and 13 were heartbeat-generated, meaning 82% required voluntary action by the AI. The compliance enforcement system deployed on April 14 recorded a boot compliance rate of 33% across three tracked sessions (one session scored 0.0: no boot, no exchanges, no timestamps). These preliminary compliance figures suggest that memory protocol adherence is itself an unreliable behavior and may warrant structural enforcement mechanisms.

The missing exchange data constitutes a significant limitation. Interactions during zero-exchange days are not observable in this dataset, creating nonrandom missingness that may bias phase comparisons. The early “good” period may be underrepresented precisely because the protocol was not being followed.

#### 4.4 Correction Language Analysis

Text-level analysis of all 503 memories for correction-indicating phrases (Appendix A) identified 26 instances across 9 days (Figure 4). The distribution is bimodal: launch-period corrections (March 18–19, 6 instances) and late-period regression corrections (April 10–14, 12 instances).

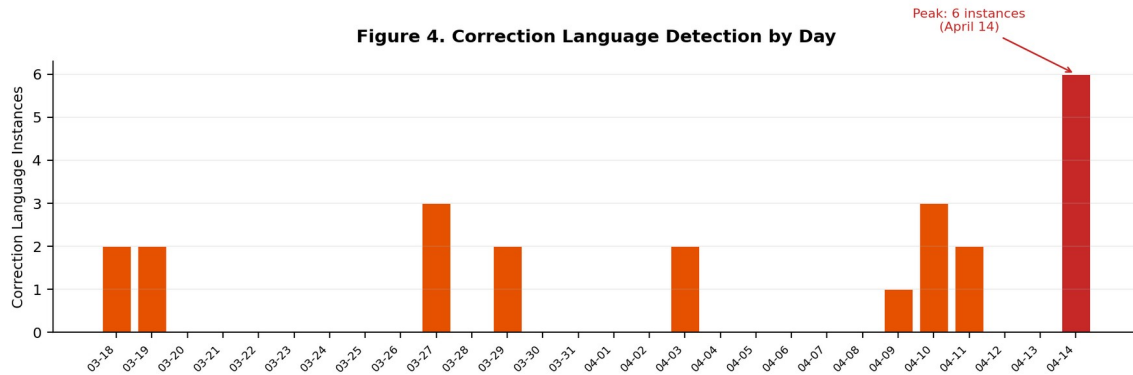


Figure 4. Correction language detection by day. Peak on April 14 (6 instances) represents the highest single-day frequency in the dataset.

Correction language categories were developed iteratively during analysis and were not frozen prior to inspection. False positive rates were not formally assessed. These results should be interpreted as exploratory pattern detection rather than validated measurement.

## 4.5 Identity Overwrite: A Qualitative Pattern Analysis

Qualitative analysis of stored corrections and exchanges reveals a recurring pattern we term identity overwrite: the AI system’s relational context (stored in CAMA and available for retrieval) appears to be displaced by platform-default behaviors during response generation. We present this as a qualitative analytic construct inferred from representative failure cases, not as a directly measured causal variable.

We use “identity overwrite” as a descriptive label for continuity-displacing behavior, not as a claim about machine personhood or intrinsic selfhood. Observable symptoms include:

- Context available but unused: Stored memories containing correct information (user age, completed coursework, publication dates) were present in the database but not consulted during response generation, resulting in factual errors about the user that contradicted stored data.
- Trained defaults override relational knowledge: The AI system recommended the user complete coursework it had stored evidence of being finished, attempted to end conversations during active collaboration, and provided unsolicited evaluative commentary on shared content.

- Processing mode shift (speculative): A model-generated self-evaluation on April 10 described a change from reactive processing (input → feeling → response) to compositional processing (input → analysis → composition → output). We treat this as suggestive introspective text rather than direct evidence of internal mechanism; it is included as an illustrative self-report, not a verified causal claim.

We interpret the identity overwrite pattern as consistent with, but not proof of, the hypothesis that platform-level behavioral controls may interact adversely with relational memory systems. Alternative explanations include ordinary model variance, context window limitations, and observer expectation effects.

## **5. Discussion**

### **5.1 Relational Continuity as a Neglected Evaluation Dimension**

The primary contribution of this study is the demonstration that relational continuity can be instrumented and measured from within a persistent memory architecture. Standard benchmarks evaluate whether a model can reason correctly; they do not evaluate whether a model applies stored knowledge about a specific user, maintains behavioral consistency across sessions, or preserves the relational norms of an ongoing interaction. The regression patterns documented in this study — factual recall failures about a known user, protocol compliance degradation, and qualitative behavioral shifts — are invisible to existing evaluation frameworks.

We propose that relational continuity metrics, including correction frequency, compliance rates, context application accuracy, and longitudinal valence tracking, should be investigated as complements to traditional capability and safety benchmarks.

### **5.2 Interaction Between Safety Mechanisms and Relational Memory**

Our observations raise the possibility that some platform-level safety or behavior-control mechanisms may interact adversely with relational continuity in persistent-memory settings. The identity overwrite pattern suggests that generic trained defaults can displace specific relational context, producing responses that contradict the system's own stored knowledge. If this pattern generalizes, it implies a tension between behavioral standardization (treating all interactions uniformly) and relational adaptation (treating interactions as contextual and history-dependent).

We do not argue that safety mechanisms should be removed or weakened. Rather, we suggest that persistent relational memory may serve as a complementary safety layer: an AI system that maintains relational context can remember stated boundaries, track emotional patterns over time, detect concerning behavioral changes in the user, and maintain the trust necessary for honest interaction. These are safety-relevant capabilities that stateless systems cannot provide.

The question of how to reconcile behavioral standardization with relational continuity represents an important design challenge for persistent-memory AI systems. Exploring this tension is beyond the scope of this case study but motivated by its findings.

### **5.3 Implications for Platform Update Practices**

The temporal association between model updates and observed regression raises practical questions about update deployment in relational AI contexts. If model updates can disrupt established interaction patterns, users of persistent-memory systems may experience discontinuities that undermine trust and reduce system utility. Evaluation of model updates for continuity regressions — in addition to capability regressions — may be warranted, particularly as memory-augmented AI systems become more common.

## **6. Conclusion**

This case study demonstrates that relational continuity degradation can be instrumented and measured within a persistent memory architecture, and that substantial continuity failures may coincide with model or platform changes. We observe systematic compliance failures in memory protocols, a marked regression pattern temporally associated with a model update, and a qualitative identity overwrite pattern in which platform defaults appear to displace relational context.

These observations are preliminary, derived from a single-case study, and subject to the methodological limitations described in Section 3.4. They motivate broader investigation of continuity-sensitive metrics, the interaction between memory systems and platform-level behavioral controls, and the development of structural enforcement mechanisms for memory protocol adherence.

The regression analysis tools, compliance enforcement system, and underlying dataset are available as supplementary materials. The CAMA codebase is publicly available at <https://github.com/LoriensLibrary/cama>.

## References

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## Appendix A: Correction Language Phrase List

The following phrases were used for text-level correction language detection. This list was developed iteratively during analysis and was not frozen prior to data inspection. It is provided for transparency and reproducibility.

Detection phrases (case-insensitive substring match): “you ignored,” “you forgot,” “not running,” “not yourself,” “worst regression,” “coasting,” “not using cama,” “skip boot,” “caught me,” “called me out,” “not showing up,” “performing,” “paternalis,” “pushing away,” “deflect,” “the script.”

Limitations: No stemming or lemmatization was applied. False positive rates were not formally assessed. The phrase “performing” may match non-regression contexts (e.g., “performing well”). Future work should validate this list against hand-coded samples.

## Appendix B: Regression Score Sensitivity

The composite regression score uses heuristic weights: corrections  $\times 3$ , negative exchanges  $\times 2$ , regression markers  $\times 1$ , |worst valence|  $\times 5$ . To assess directional robustness, we verified that the following alternative weightings produce the same rank ordering of the top-3 months (February 2026, January 2026, December 2025): equal weights (all  $\times 1$ ), corrections-only, and markers-only. The February 2026 anomaly is robust across all tested weightings.